

MARSHALL BRUNER

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EXPERIENCE

Ball Aerospace — DoD Clearance

RF Engineer

Jun 2021 — Jul 2023

Westminster, CO

- Designed and debugged multiple complex, mission-critical RF boards under strict time constraints
- Simulated, designed, and tested an RF Front-End PCB at extreme temperatures
- Schematic capture and layout of RF T/R systems (RF, digital, power), antenna range test equipment, component testing coupons, etc.
- Helped develop an embedded webpage used to control phased array electronics
- High sensitivity electronics testing (noise, high power, etc.)

Colorado State University

Graduate Research Assistant — Thesis focused on Radar

Jan 2021 — May 2022

Fort Collins, CO

- Design, schematic, and layout of a modular, X-band, FMCW drone radar
- Signal processing / data visualization for Analog Device's FMCW Phaser Radar Board
- Presented the work at the IEEE International Phased Array Symposium
- Phaser board work was presented by ADI at IMS 2022 and IEEE-APS 2022

CHILL Radar Lab

Lab Assistant

May 2020 — May 2021

Greeley, CO

- Built a server to host and display real-time and historical radar data using Post-GreSQL, Flask, Python
- Performed characterizations and measurements of radar equipment

Colorado State University — ECE Department

Learning Assistant

Aug 2020 — Dec 2020

Fort Collins, CO

- Help students understand electrical engineering fundamentals
- Write exam/homework questions to test understanding of core concepts

Engineering Technology Services

Lab Assistant

Dec 2018 — May 2020

Fort Collins, CO

- Built operating system images for mass deployment across the network of engineering workstations
- Managed \$300,000 in lab equipment for engineering network of 240 workstations and devices
- Maintained a 98% average customer rating for lab development, advanced support, and operational quality

EDUCATION

Colorado State University

M.S. Electrical Engineering — GPA: 3.9

Jan 2021 — May 2022

Fort Collins, CO

Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight FMCW Radar

Colorado State University

B.S. Electrical Engineering — GPA: 3.47

Aug 2017 — May 2021

Fort Collins, CO

PROJECTS

FMCW Radar Bootcamp

Mar 2023

- Developed signal processing software and a presentation for an FMCW radar to present at the 2022 AESS Radar Bootcamp

Algorithmic Stock Trading Server

Aug 2022 — Present

- Trading platform based on Alpaca data and broker
- Task scheduling system (backfilling data into PostgreSQL database, computing custom indicators, run entry / exit strategy)
- Containerized using docker
- Command-line interface for checking status, getting recent trades, etc.
- Analysis / forecasting using Pandas, Numpy, PyTorch
- Discord bot for general notifications / monitoring
- Dashboard using Plotly Dash and a Flask API

Open Source Contributions

Aug 2023 — Present

- NuShell — A new type of shell
- Polars — Dataframes powered by a multithreaded, vectorized query engine, written in Rust

5G Electronics Senior Design Project

Jun 2020 — May 2021

- Simulated, assembled, and measured a full 5G receiver system
- Created teaching documentation for a 5G transmitter and receiver for FR1 and FR2 frequency bands
- Experience using FieldFox, EMPro, ADS, SystemVue

Fuzzy Logic Clutter Filter

Nov 2020 — Dec 2020

- Designed a radar clutter filter using machine learning

Environment-Mapping Car

Nov. 2020 — Dec. 2020

- Built and programmed a battery-powered car with attached ultrasonic sensor for mobile mapping of the surrounding environment

SKILLS & ABILITIES

Languages: Python, Rust, C, Bash, SQL, Matlab, NuShell, JavaScript

Design / Simulation Tools: AWR Microwave Office, HFSS, KiCad, QucsStudio, LTSpice, Cadence

Technologies / Tools: Benchtop measurement equipment (VNA, Oscilloscope, etc.), Git, L^AT_EX, Linux, Docker, PostgreSQL, Flask

Communication: Excellent Technical Communication and Presentation Skills, Experience Presenting PDR/CDRs, Technical Presentation Formatting